Kidney stones are a common ailment affecting approximately 10% of adults in the United States. They form when solutes precipitate out of solution as crystals in the urinary tract, and they can cause severe pain in the side, back, abdomen, and groin. Individuals who have been previously diagnosed with kidney stones have an increased probability of developing new stones relative to unaffected individuals. Different measures may help prevent the formation of different kinds of stones, so analysis of the composition of stones that have been passed or removed can aid in preventing recurrence. Stones can be ground into fine powders, dissolved in a small amount of solvent, and analyzed by infrared (IR) spectroscopy, as shown in Figure 1.

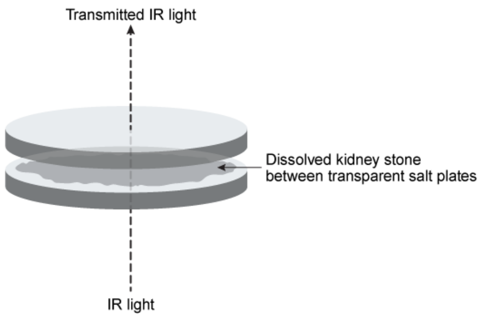

Figure 1 Schematic of kidney stone analysis by IR spectroscopy

IR analysis of kidney stones from 50 individuals revealed the percentage of stones that contain each component, shown in Table 1 along with solubility data.

Table 1 Kidney Stone Composition Parameters

Kidney stone component	Frequency (% stones analyzed)	K_{sp} at pH 7 (25°C)	Solubility change with increasing pH
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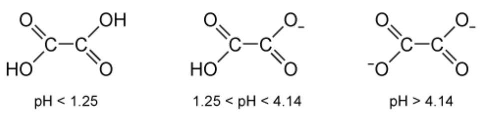


Figure 2 Structure of oxalic acid and its associated anions with increasing pH

Potassium citrate alkalinizes the urine, potentially causing a decrease in oxalate solubility and the formation of more crystals. However, potassium citrate can also react with calcium oxalate according to the unbalanced equation shown in Reaction 1:

$$CaC_2O_4 + K_3(C_6H_5O_7) \rightarrow Ca_3(C_6H_5O_7)_2 + K_2C_2O_4$$

Reaction 1

Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

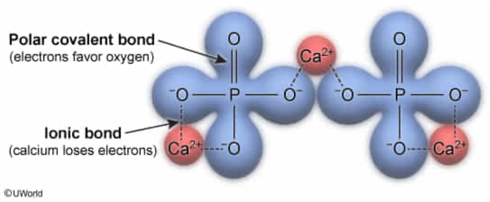
Which of the following bond types are found in the calcium phosphate present in 35% of kidney stones?

I. Ionic
II. Polar covalent
III. Nonpolar covalent

- A. I and II only [70%]
- B. I and III only [17%]
- C. II and III only [4%]
- D. I, II, and III [6%]

Correct
70% answered correctly

Explanation:



Different atoms have different affinities for electrons, expressed as electronegativity. The **type of bond** that forms between two atoms depends on the **electronegativity difference** between the atoms. When atoms have an electronegativity difference greater than about 1.7, **electrons are fully transferred** from the less electronegative atom to the more electronegative atom. This process forms charged particles called ions and creates an ionic bond. Metals such as the calcium in calcium phosphate have low electronegativity and almost always participate in ionic bonds **(Number I)**.

Atoms with an electronegativity difference between about 0.5 and 1.7 do not fully transfer electrons but instead **share electrons unequally**. The atom with the higher electronegativity gains a partial negative charge as it exerts a stronger pull on the electrons whereas the less electronegative atom gains a partial positive charge. This separation of charge forms an electric dipole, and bonds of this nature are called **polar covalent bonds**. The phosphate ion is composed of a phosphorus atom covalently bonded to several oxygen atoms. Oxygen is the second-most electronegative atom on the periodic table, and the covalent bonds that it forms are almost always polar, as in the case of phosphate **(Number II)**.

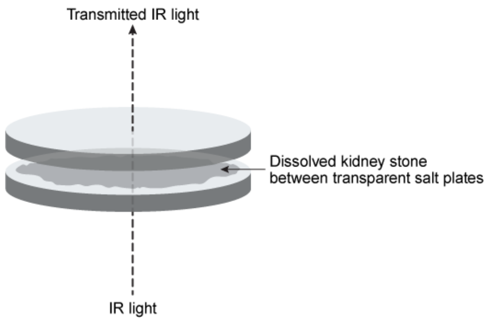
(Number III) Nonpolar covalent bonds form between atoms with a difference in electronegativity no greater than 0.5. These bonds involve **equal sharing of electrons**, and they frequently form between two atoms of the same type (eg, carbon-carbon bond). Carbon-hydrogen bonds are also considered nonpolar because both atoms have similar electronegativities. No such bonds exist in calcium phosphate.

Educational objective:
Bonds between atoms are classified generally as ionic, polar covalent, or nonpolar covalent. Ionic bonds involve complete transfer of electrons from an atom with low electronegativity to an atom with high electronegativity. Covalent bonds involve electron sharing, and are considered polar if the sharing is unequal but nonpolar if the sharing is equal.

Time spent:0 sec
Subject:
General Chemistry

QID:401285
System:
Interactions of Chemical Substances

Kidney stones are a common ailment affecting approximately 10% of adults in the United States. They form when solutes precipitate out of solution as crystals in the urinary tract, and they can cause severe pain in the side, back, abdomen, and groin. Individuals who have been previously diagnosed with kidney stones have an increased probability of developing new stones relative to unaffected individuals. Different measures may help prevent the formation of different kinds of stones, so analysis of the composition of stones that have been passed or removed can aid in preventing recurrence. Stones can be ground into fine powders, dissolved in a small amount of solvent, and analyzed by infrared (IR) spectroscopy, as shown in Figure 1.



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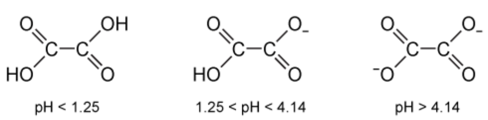

Figure 2 Structure of oxalic acid and its associated anions with increasing pH

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Reaction 1

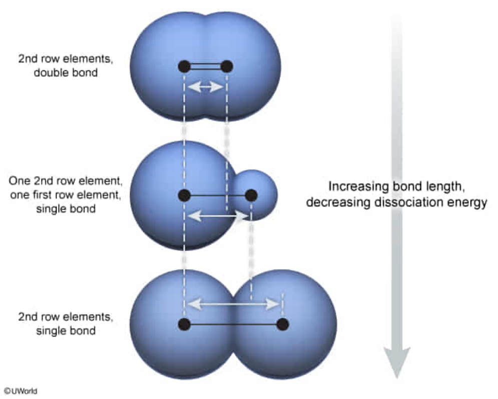
Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

Consider the trends in atomic radii on the periodic table. The bond in citrate that is the longest and weakest with the lowest bond dissociation energy is the:

- A. C–H bond. [40%]
- B. C=O bond. [2%]
- C. C–C bond. [55%]
- D. The bond length and strength cannot be determined without more information. [1%]

Correct
54% answered correctly

Explanation:



Bond dissociation energy is the energy required to **break a chemical bond**; it is related to the **bond length**, or the distance between the nuclei of two bonded atoms. **Shorter bonds are stronger** and require more energy to break. Atoms with small **atomic radii** can form short, strong bonds whereas atoms with larger radii form longer, weaker bonds. Each subsequent row on the periodic table indicates an additional electron shell, and therefore a larger atomic radius, so atoms in the first row will have smaller atomic radii than atoms in the second row, and so on. Double bonds require more energy to break than single bonds.

Citrate includes bonds between carbon and carbon, carbon and oxygen, and carbon and hydrogen. Of the choices given, the C–C bond is the only single bond that does not involve a first row atom (hydrogen). Therefore, the C–C bond will be the longest and require the least amount of energy to break.

(Choice A) This option includes hydrogen (a first row atom), which forms short, strong bonds.

(Choice B) C–O single bonds are of approximately the same strength as C–C single bonds (both C and O are second row atoms). However, C=O double bonds are much shorter and stronger.

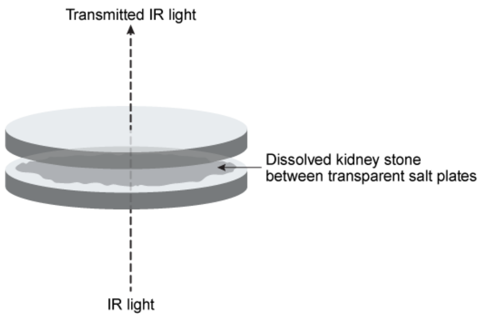
(Choice D) Bond length and strength can be estimated based on trends in atomic radii on the periodic table. Therefore, there is sufficient information to determine which bond in citrate is the longest and weakest.

Educational objective:
The energy required to break a bond, known as the bond dissociation energy, is related to the bond length. Atoms with smaller radii (near the top of the periodic table) tend to form shorter, stronger bonds than atoms with larger radii.

Time spent:0 sec
Subject:
General Chemistry

QID:401286
System:
Interactions of Chemical Substances

Kidney stones are a common ailment affecting approximately 10% of adults in the United States. They form when solutes precipitate out of solution as crystals in the urinary tract, and they can cause severe pain in the side, back, abdomen, and groin. Individuals who have been previously diagnosed with kidney stones have an increased probability of developing new stones relative to unaffected individuals. Different measures may help prevent the formation of different kinds of stones, so analysis of the composition of stones that have been passed or removed can aid in preventing recurrence. Stones can be ground into fine powders, dissolved in a small amount of solvent, and analyzed by infrared (IR) spectroscopy, as shown in Figure 1.



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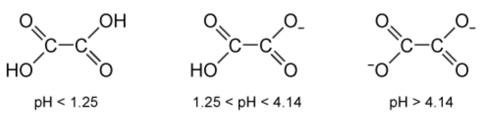


Figure 2 Structure of oxalic acid and its associated anions with increasing pH

Potassium citrate alkalinizes the urine, potentially causing a decrease in oxalate solubility and the formation of more crystals. However, potassium citrate can also react with calcium oxalate according to the unbalanced equation shown in Reaction 1:

$$CaC_2O_4 + K_3(C_6H_5O_7) \rightarrow Ca_3(C_6H_5O_7)_2 + K_2C_2O_4$$

Reaction 1

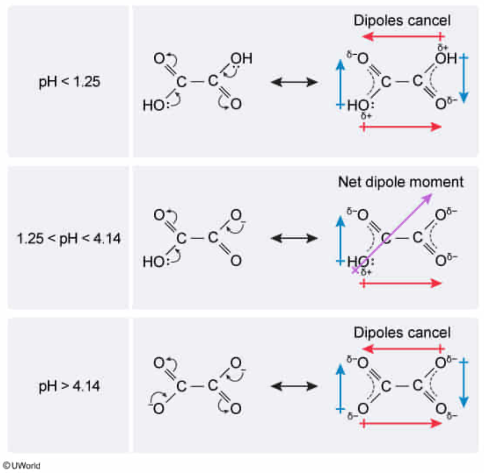
Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

At which pH would oxalic acid adopt the greatest net dipole moment?

- A. 1 [10%]
- ✔ B. 3 [69%]
- C. 5 [7%]
- D. 7 [11%]

Correct
69% answered correctly

Explanation:



A dipole moment occurs when positive charge accumulates at one end of a molecule and negative charge accumulates at the other. The greater the **difference in charge** at each end of the molecule, the greater the dipole moment. A molecule may have multiple polar bonds, each with its own dipole moment. If the individual moments all point in the same direction, the overall moment is increased. If individual moments point in opposite directions, they cancel each other and the molecule as a whole has no dipole. For this reason, **symmetrical molecules** generally **do not have net dipole moments** and are considered nonpolar.

As shown in Figure 2, oxalic acid can take on different forms depending on the pH to which it is exposed. When the pH < 1.25, the molecule is fully protonated. Resonance forms show that individual regions of this molecule have dipole moments. However, this form of the molecule is symmetrical so the dipoles cancel, giving no net dipole moment. Similarly, while the molecule is fully deprotonated at a pH > 4.14, it is still symmetrical and has no net dipole moment in this form either.

When the pH is between 1.25 and 4.14, the molecule is **half protonated** and is **no longer symmetrical**. In this form, it has a dipole moment. Therefore, oxalic acid has the greatest net dipole moment at a pH of 3.

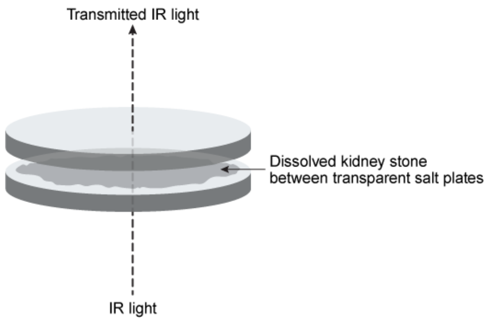
(Choices A, C, and D) At a pH below 1.25 or above 4.14, the molecule becomes symmetrical and all dipoles cancel.

Educational objective:
Molecules have a net dipole moment (separation of charge) when the individual dipoles within it do not cancel each other. Symmetrical molecules typically do not have net dipole moments. Protonation and deprotonation can change whether a molecule is symmetrical and can induce or remove a dipole moment.

Time spent:0 sec
Subject:
General Chemistry

QID:401287
System:
Interactions of Chemical Substances

Kidney stones are a common ailment affecting approximately 10% of adults in the United States. They form when solutes precipitate out of solution as crystals in the urinary tract, and they can cause severe pain in the side, back, abdomen, and groin. Individuals who have been previously diagnosed with kidney stones have an increased probability of developing new stones relative to unaffected individuals. Different measures may help prevent the formation of different kinds of stones, so analysis of the composition of stones that have been passed or removed can aid in preventing recurrence. Stones can be ground into fine powders, dissolved in a small amount of solvent, and analyzed by infrared (IR) spectroscopy, as shown in Figure 1.



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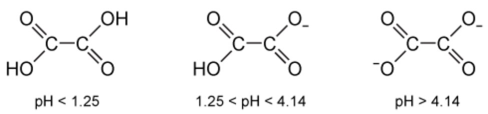


Figure 2 Structure of oxalic acid and its associated anions with increasing pH

Potassium citrate alkalinizes the urine, potentially causing a decrease in oxalate solubility and the formation of more crystals. However, potassium citrate can also react with calcium oxalate according to the unbalanced equation shown in Reaction 1:

$$CaC_2O_4 + K_3(C_6H_5O_7) \rightarrow Ca_3(C_6H_5O_7)_2 + K_2C_2O_4$$

Reaction 1

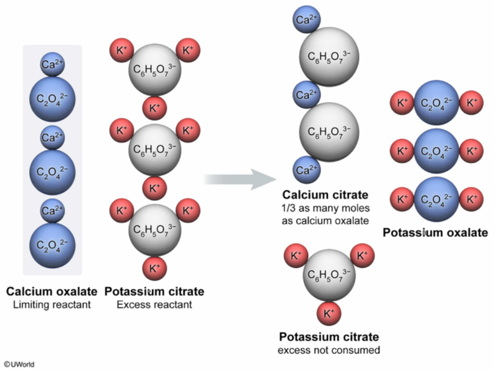
Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

Given the unbalanced equation (Reaction 1) and the molecular weight of calcium citrate (498.5 ng/nmol), if 15 nmol of calcium oxalate is mixed with 15 nmol of potassium citrate, what is the approximate theoretical yield of calcium citrate?

- A. 1,250 ng [4%]
- B. 2,500 ng [55%]
- C. 3,750 ng [22%]
- D. 7,500 ng [16%]

Correct
56% answered correctly

Explanation:



Theoretical yield is the **maximum amount of product** that can form during a reaction based on the limiting reactant, which is the reactant that will be **entirely consumed** if the reaction goes to completion. The limiting reactant can be determined from the **balanced chemical equation** and the number of moles of each reactant.

The passage gives the unbalanced equation for the reaction between calcium oxalate and potassium citrate. The equation must be balanced to determine the limiting reactant. The balanced equation is:

$$3\ CaC_2O_4 + 2\ K_3C_6H_5O_7 \rightarrow Ca_3(C_6H_5O_7)_2 + 3\ K_2C_2O_4$$

The equation shows that the formation of 1 mole of calcium citrate requires 3 moles of calcium oxalate but only 2 moles of potassium citrate. The question states that 15 nmol of each reactant were mixed together. As such, a quantity of 15 nmol of calcium oxalate can form 5 nmol of calcium citrate:

$$15\ \cancel{\text{nmol CaC}_2\text{O}_4} \times \frac{1\ \text{nmol Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2}{3\ \cancel{\text{nmol CaC}_2\text{O}_4}} = 5\ \text{nmol Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2$$

However, 15 nmol of potassium citrate can produce 7.5 nmol of calcium citrate:

$$15\ \cancel{\text{nmol K}_3\text{C}_6\text{H}_5\text{O}_7} \times \frac{1\ \text{nmol Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2}{2\ \cancel{\text{nmol K}_3\text{C}_6\text{H}_5\text{O}_7}} = 7.5\ \text{nmol Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2$$

The limiting reactant is the one that yields the least amount of product. In this case, calcium oxalate is the limiting reactant. The maximum amount of calcium citrate formed in ng from the reaction can be calculated by converting nmol to ng using the molecular weight of calcium citrate (≈ 500 ng/nmol):

$$5\ \cancel{\text{nmol}} \times \frac{500\ \text{ng}}{\cancel{\text{nmol}}} = 2{,}500\ \text{ng Ca}_3(\text{C}_6\text{H}_5\text{O}_7)_2$$

Therefore, the approximate theoretical yield is 2,500 ng of calcium citrate.

(Choice A) This value is half of the correct answer. Concentration, not total number of moles, must be divided by 2 when equal volumes of different solutions are mixed.

(Choice C) If calcium oxalate was in excess, potassium citrate would be the limiting reactant, and 7.5 nmol or 3,750 ng of calcium citrate would be the theoretical yield.

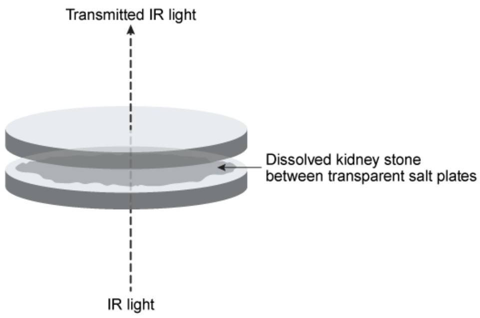
(Choice D) Using a 1:1 molar ratio for the reactants and products (from the unbalanced reaction) incorrectly determines that 15 nmol of each reactant would produce 15 nmol of calcium citrate with a mass of 7,500 ng. However, the reaction must be balanced to achieve the correct molar ratios.

Educational objective:
Theoretical yield is the maximum amount of product that can be formed in a reaction. It is governed by the limiting reactant, which is determined by the balanced equation and the molar amounts of each reactant available.

Time spent:0 sec
Subject:
General Chemistry

QID:401288
System:
Interactions of Chemical Substances

Kidney stones are a common ailment affecting approximately 10% of adults in the United States. They form when solutes precipitate out of solution as crystals in the urinary tract, and they can cause severe pain in the side, back, abdomen, and groin. Individuals who have been previously diagnosed with kidney stones have an increased probability of developing new stones relative to unaffected individuals. Different measures may help prevent the formation of different kinds of stones, so analysis of the composition of stones that have been passed or removed can aid in preventing recurrence. Stones can be ground into fine powders, dissolved in a small amount of solvent, and analyzed by infrared (IR) spectroscopy, as shown in Figure 1.



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Some studies indicate that potassium citrate, taken orally, may prevent the formation of calcium oxalate crystals, the most abundant component of kidney stones. Oxalic acid, shown in Figure 2, is significantly more soluble than calcium oxalate.

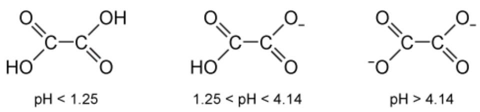


Figure 2 Structure of oxalic acid and its associated anions with increasing pH

Potassium citrate alkalinizes the urine, potentially causing a decrease in oxalate solubility and the formation of more crystals. However, potassium citrate can also react with calcium oxalate according to the unbalanced equation shown in Reaction 1:

$$CaC_2O_4 + K_3(C_6H_5O_7) \rightarrow Ca_3(C_6H_5O_7)_2 + K_2C_2O_4$$

Reaction 1

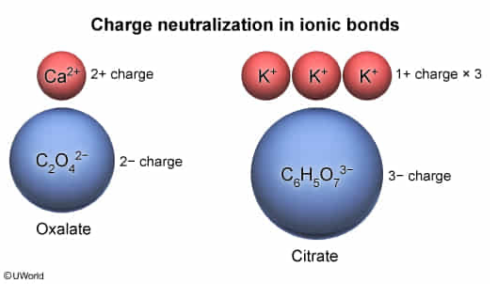
Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

Based on Reaction 1 in the passage, what are the charges of the oxalate and citrate ions, respectively?

- A. −1, −3 [5%]
- B. −2, −2 [4%]
- ✔ C. −2, −3 [78%]
- D. −3, −2 [11%]

Correct
77% answered correctly

Explanation:



Ionic bonds form between **positively charged** and **negatively charged** ions such that both positive and negative **charges are neutralized**. If the charges of the cations are known, the charges of the anions can be determined by balancing the charges.

The arrangement of the periodic table indicates charges that atoms are likely to adopt. Metals in the first column of the periodic table, including potassium, ionize by losing one electron; potassium has a charge of +1. Metals in the second column, such as calcium, lose two electrons and have a +2 charge. The reaction in the passage shows that one calcium ion is neutralized by one oxalate ion, and three potassium ions are neutralized by one citrate ion. Therefore, oxalate must have a charge of −2 and citrate a charge of −3 to neutralize their respective cations.

(Choice A) As a second column element, calcium acquires a +2 charge. Therefore, a single calcium cation can be neutralized by a single anion such as oxalate only if that anion has a charge of −2.

(Choice B) Citrate is neutralized by three potassium ions, each of which has a +1 charge. Therefore, citrate must have a charge of −3.

(Choice D) This option reverses the charges of oxalate and citrate.

Educational objective:

Ionic bonds form when positive charges neutralize negative charges. One ion may carry multiple charges, and must be neutralized by the same number of opposite charges.

Time spent:0 sec
Subject:
General Chemistry

QID:401289
System:
Interactions of Chemical Substances

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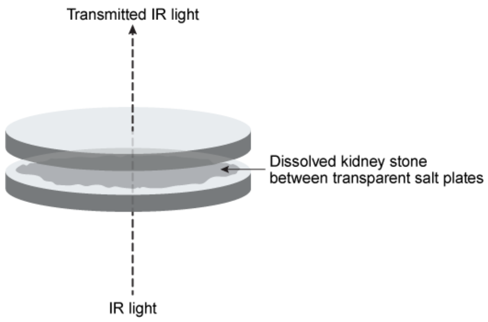

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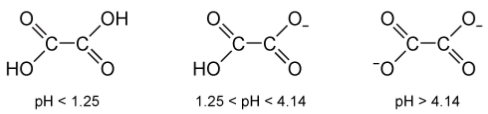


Figure 2 Structure of oxalic acid and its associated anions with increasing pH

Potassium citrate alkalinizes the urine, potentially causing a decrease in oxalate solubility and the formation of more crystals. However, potassium citrate can also react with calcium oxalate according to the unbalanced equation shown in Reaction 1:

$$CaC_2O_4 + K_3(C_6H_5O_7) \rightarrow Ca_3(C_6H_5O_7)_2 + K_2C_2O_4$$

Reaction 1

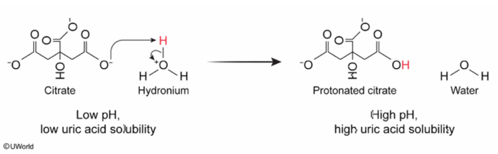
Calcium citrate and potassium oxalate are both hundreds of times more soluble than calcium oxalate, so the presence of citrate and potassium ions can help maintain calcium and oxalate ions in solution. This effect may be sufficient to overcome the decreased solubility that occurs at higher pH levels.

In addition to calcium oxalate stones, researchers might hypothesize that potassium citrate (a weak base) could reduce the risk of uric acid stones for which of the following reasons?

- A. Potassium reduces the hydroxide concentration in urine. [15%]
- B. Potassium increases the hydroxide concentration in urine. [21%]
- ✔ C. Citrate reduces the hydronium concentration in urine. [50%]
- D. Citrate increases the hydronium concentration in urine. [12%]

Correct
50% answered correctly

Explanation:



The pH of a solution is defined as the **negative logarithm** of the hydronium concentration, often expressed in a simplified manner as −log [H^+]. pH is related to the negative logarithm of hydroxide concentration (pOH) by the equation **pH = 14 − pOH**. Therefore, pH can be **increased by adding hydroxide** ions or by **removing hydronium** ions.

The passage states that citrate alkalinizes (increases the pH of) the urine, and Table 1 indicates that uric acid becomes more soluble with increasing pH. The more soluble a compound is, the less likely it is to precipitate and form kidney stones. Citrate is a weak base and therefore a proton acceptor. It can alkalinize a solution by removing protons from hydronium ions, converting those ions to water and reducing the total hydronium concentration. Therefore, citrate may increase the solubility of uric acid by reducing the hydronium concentration and may decrease the risk of uric acid kidney stones.

(Choice A) Reducing the hydroxide concentration would decrease the pH.

(Choice B) Increasing the hydroxide concentration requires deprotonation of water to form hydroxide. Potassium is positively charged and would repel positively charged protons, and so be unable to deprotonate water.

(Choice D) Citrate decreases the hydronium concentration. Increasing the hydronium concentration would lower the pH and decrease uric acid solubility.

Educational objective:

The pH of a solution is equal to the negative logarithm of the hydronium ion concentration. It is related to hydroxide concentration by pH = 14 − pOH and can be increased by adding hydroxide ions or removing hydronium ions.

Time spent:0 sec
Subject:
General Chemistry

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System:
Interactions of Chemical Substances